

ScriptDole

“Social Basic Income on the Radix Network”

Whitepaper V. 0.1 by VandyILL

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Scripto: RadixDLT’s asset-oriented programming language designed to work with the Radix Engine for fast, secure and scalable DeFi applications.

Dole: 1) Something portioned out bit by bit; 2) a benefit paid by the government to the unemployed.

Abstract:

ScriptDole is an unsecured currency¹ minted on the RadixDLT network and distributed as a “social token” at a fixed rate of 100 per month to each registered user. The goal of the ScriptDole currency system is to implement a form of universal basic income with value set by the market and social consensus. If this goal is met then hopefully it changes the conversation of how certain goods and services should be priced and whether the ScriptDole basic income is sufficient for survival and opportunity in the modern economy.

Movements, not flash mobs.

¹ Unsecured Currency means that there are no assets backing the tokens. The ScriptDole component mints tokens at a fixed rate per registered user but offers no redemption option. The ScriptDole does this at no cost to the recipient. Any idea, acceptance, or adoption of the currency is entirely driven by society and markets.

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I. Introduction

Technological developments have dramatically lowered the cost for developing and implementing financial tools.

With this newfound agency it is imperative to ask what type of world we want and work backwards to design financial systems and market forces that will achieve those ends.

With end goals in mind this paper aims to push forward with a new financial tool, a form of Social Basic Income (SBI), and to contribute to the forward momentum of the cryptocurrency industry and broader movement of alternative finance.

Specifically, this paper lays out the case for distributing an unbacked, unpegged digital currency every month to every registered participant as a form of universal basic income.

Instead of using the term Universal Basic Income, which is already a debated and tested policy, we are opting to use the term Social Basic Income as the value of the currency will be decided by society and the market, not central entities such as governments.

This currency, ScryptDole, will be implemented on the Radix decentralized ledger network, and utilize a 'blueprint' which can be reused by other groups looking to implement their own variations of an SBI. This can be done at any level or for any community, ranging from a global community like ScryptDole, to a local village or online groups. Sub-global implementations of SBI can also be a strong force for decentralized identity when coupled with Decentralized Autonomous Organizations (DAO's) and non-transferable Radix badges and rights or recognition.

This paper describes the basics of ScryptDole (Section II), a rather simple system with potentially far ranging impact, then reviews a short, real world case study about Bitcoin (Section III). After introducing these topics there is a brief primer about Cryptocurrency as a subset of the Alternative Finance movement (Sections IV). This quick introduction should garner interest into how a currency can improve economies even if it is not backed by any specific asset.

With some historic proof in place, plus context for the Alternative Finance movement, the next sections cover technical aspects of cryptocurrency and its history, concluding the topic with a case for why Radix is the best network for this project (Sections V-VII.)

The next two sections cover new topics that are supplemental to the core of the SBI system. This includes a dive into SBI's, DAO's and Radix badges as laying the foundation for decentralized identity (Section VIII.) This is followed by discussion of a new concept, "Confidence Farming," which is a new tool for ensuring currency liquidity and hedging against impermanent loss (Section IX.)

The paper then covers several examples of how a social currency can gain traction, and begins the conversation on how society may organize or empower a UBI social currency (Section X.)

The remaining sections are for work that needs to be done, including addressing problems and critiques (Section XI), learning from competitors (Section XII), and planning next steps to bring the system to market.

At the end of this paper you should have a basic understanding of how social basic income has merit, how it will be implemented, and potentially view ScriptDole as an asset that does not need to be pegged to another currency. Instead, if Successful ScriptDole may actually emerge as a preferable system for pricing some goods and services in the real world as the origination and numeration of the money is aligned with unique values compared to existing currency systems, and even other universal basic income systems.

As a final note, this paper attempts to balance the needs of readers who are foreign to cryptocurrency, and readers who are new to alternative finance. Due to these constraints only a top level view is taken and this is not intended to be a disservice to the complexity of these topics in their own right.

II. ScriptDole

The goal of ScriptDole is to introduce a piece to the global financial system that contributes to a more equitable economy.

ScriptDole consists of two parts, a ScriptDole component (the Radix network's version of a smart contract), and a ScriptDole currency, or 'token.'

The function of the ScriptDole component is to verify that a real person is registered and willing to receive a monthly allotment of ScriptDole and to mint and deliver the allotment each month. ScriptDole will utilize Radix's Instapass service, a Know Your Customer (KYC) solution to verify that users are unique and not double registering to increase their monthly allotment. Meanwhile, the component will dole out 100 ScriptDole tokens each month to each registered user.

The ScriptDole component will also have the capability to mint new tokens necessary to pay transaction fees on the Radix network in the network's native token, XRD. With "atomic composability" - the ability to settle multi-step transactions as one transaction - the Radix network will enable the following steps to happen in one transaction:

- 1) Check the price of XRD/SD
- 2) Mint new SD
- 3) Swap SD for XRD
- 4) Distribute remaining SD to users as their monthly dole, and/or to stakers on a "Confidence Farm"

5) Pay transaction fees with XRD from step 3

In final form, there may be several extra steps related to this process if the Confidence Farm system is implemented, but the five steps outlined above are the key components for a functioning SBI.

Neither the ScriptDole component, nor the ScriptDole Foundation will offer any redemption options for ScriptDole; nor will they back the currency by any assets; nor will there be any attempt to peg the currency to a specific price.

Any value placed on ScriptDole will be entirely determined by the community and via any mechanisms employed by the community such as decentralized exchanges or new innovations. Taking a long view, if ScriptDole is accepted into society it may become a constant social movement or issue, akin to a minimum wage, pensions, union rights and labor laws, but without government intervention and no explicit obligation on any unwilling party like a tax for a traditional universal basic income would create. Every market participant will decide what, if any value they want to place in the currency.

In regards to value, the concept of Confidence Farming is an exception. Confidence Farming is a means to create a more stable currency with high liquidity and lessen the possibility of impermanent loss for liquidity providers working with Automated Market Makers. Confidence Farming is not a necessary component of ScriptDole, but it is an option that could make the currency more stable (but not necessarily pegged to a specific asset.) Further, Confidence Farming could lure non-traditional or new crypto investors such as charities who are seeking safer investments with more modest returns, or classes of investors who are looking to make an impact with their investing² which is a growing movement among many investors.

Given the current trajectory of the cryptocurrency movement this approach is likely to draw sharp criticism from the community for reasons such as a supply which could increase in perpetuity. However, this paper dedicates time to discuss Bitcoin's history and tokenomics in acknowledgement of the fact that many people who read this paper may be severely off-put by certain mechanics. Hopefully the time spent on this will add a new perspective that may serve some audiences well.

When reconsidering these issues from a new perspective it is obvious that there is a large amount of white space for new approaches in which cryptocurrency can flourish. For example, many people have bought into the cryptocurrency movement as a hedge against inflation and seek projects with a finite supply of tokens. Despite this, inflation is not the issue. The issue is scarcity and constraints and managing the stock and flow of the currency in circulation.

² <https://www.vox.com/recode/23141993/anticapitalist-investing-rich-heirs-explainer>

The distribution of ScriptDole is pegged to actual figures - time, and number of users - so there is some form of scarcity. Further, over time as the supply in circulation increases, the Stock/Flow ratio will continue to grow in size and even eclipse traditional assets like gold.

Hence, the question for cryptocurrency and alternative finance advocates to ask is whether a currency system fixed to finite numbers, or secured by an asset, or a currency system fixed to real world matters such as population is more equitable, productive, or practical. Forward looking ideas rather than just maintenance of the status quo should be the aim of our financial system and hopefully this paper and ScriptDole further conversations in that space.

The impact of the ScriptDole component will also expand beyond just ScriptDole as it will be instantiated from a blueprint on the Radix ledger. This technological infrastructure can be reused by any other party seeking to make similar or customized SBI systems. The ScriptDole Foundation's hope is that this easy to implement system sparks more development and experimentation in the alternative finance space.

Additionally, the implementation of SBI systems at local levels and with informal groups could work in conjunction with Radix's badge resource as a means to start building decentralized communities and identity³. The SBI dispersals means there is an incentive to only grant the requisite SBT to members of the community who are deserving of the SBI dispersal, as allowing too many fake, unqualified or unverified members would dilute the value of other members' allotment from the SBI system. This alignment of incentives could be the ignition necessary to launch a truly grassroots and decentralized identity system. Such efforts could even displace the need for Instapass for Scriptdole or another global SBI token.

III. El Zonte: A Case Study in Currencies & Circular Economies

Before diving into more specifics about ScriptDole it is useful to take a single case study involving Bitcoin to demonstrate, with empirical evidence, the ability of a cryptocurrency to power economic enablement even if there are no 'ramps' to a fiat currency.

Bitcoin's zealous community may be even more revolutionary than the technology itself and has conducted many social experiments which can help guide future developments in the alternative finance space.

Bitcoin had fairly negligible value for the first couple years of its existence. The first major milestone in price movement is likely the famous story of someone buying a pizza with Bitcoin, which pegged the price at about \$0.04/BTC. A miniscule fraction of its current value of approximately \$30,000/BTC.

However, an even more intriguing case study is the story of El Zonte, or "Bitcoin Beach."

³ The Badge Feature can be used to implement systems like "Souldbound Tokens" as discussed by Vitalek Buterin in a May 2022 paper.

In 2019 a nonprofit in El Zonte, a remote surfing village in El Salvador, received a stipend of Bitcoin with the condition that the money cannot be converted into fiat and must be dispersed as Bitcoin.

El Zonte is a small town, with little access to banking.

At first, Michael Peterson, the person in charge of the nonprofit disbursing Bitcoin, tried gathering the town's adults to pitch the digital currency system. Most thought it was a scam and none of them decided to participate.

Peterson then changed tactics and gave Bitcoin to the town's teenagers for cleaning up the river. Eventually, someone decided to accept bitcoin for a cup of coffee. Now, "Workers ... receive their salaries and pay bills in Bitcoin, tourists can buy pupusas with a special Bitcoin payment app, and community projects are financed with Bitcoin donations. According to Jorge Valenzuela, an upbeat 32-year-old surfing aficionado who leads the volunteers, "it has changed my town."⁴

This adoption of Bitcoin in a 'circular economy' was not dependent on any external value of Bitcoin. Simply enabling locals with a unit of account to reward work was sufficient to stimulate the economy and help participants.

Ultimately, as Bitcoin gained more traction in El Salvador, where it has even become legal tender, the unit of exchange helped connect El Zonte with the outside world. This also allowed El Zonte residents to benefit from the rising value of Bitcoin. However, the residents also suffered from the volatile nature of Bitcoin, raising the debate about how valuable it is as a tender for their economy as El Zonte and El Salvador get connected to the whole world of Bitcoin, rather than acting as the circular currency that first gained traction in El Zonte.

The key takeaways from this story are that a cryptocurrency does not need inherent value to be successful. Bitcoin could not be exchanged for anything in its infancy, or in El Zonte when it was first introduced. Despite this, in both cases it was still able to gain some social standing. This was especially crucial in El Zonte as it rewarded work that may have otherwise gone undone.

A simple unit of account dispersed in limited fashion may be sufficient to create some degree of economic improvement. This is true whether it is Bitcoin growing from nothing, to a pizza, to an international phenomenon, or for a token introduced to an isolated economy and giving rewards and organizational tools to underemployed people and unaddressed issues, like the teens and trash in El Zonte. A ledger alone may be a social tool that enables more market

participation, organization and collective action, regardless of how the units on the ledger are dispersed.

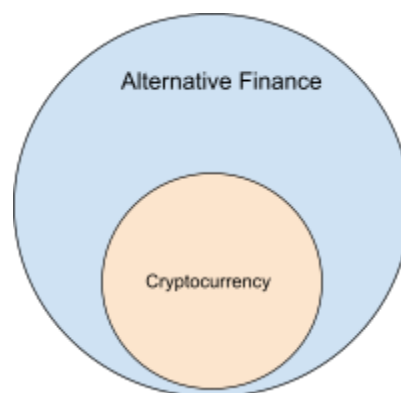
This paper will later lay out hypothetical examples of how ScryptDole may follow a path on a global scale. In the meantime this case study should serve to pique the interest of readers while working through this paper. Further, if these discussions interest you more than cryptocurrency industry focused information, then skip to the Section X: Hypothetical Value Stories.

It is also worth noting that neither ScryptDole, Bitcoin or a local implementation of an SBI system needs to be the only unit of account for a community like El Zonte. There can be multiple types of currencies for stimulating different parts of the economy or serving different social and economic purposes. Additionally, on the Radix network it should be cost efficient and frictionless to swap between payment methods.

The notion of layered SBI systems could be extremely relevant. An individual person could potentially receive ScryptDole, a global token, on top of a regional token, a national token, and a local token. Each of these could have its own value, creating different and decentralized standards for what the accumulated value of those tokens should be for someone to succeed in any individual location.

With ScryptDole defined, and an empirical use case for unsecured assets, this paper now shifts to discussing the alternative finance movement and cryptocurrency in general. These topics are specifically tailored to look at how the technology has evolved to enable something like ScryptDole.

IV. A Primer on Alternative Finance & Cryptocurrency



The Alternative Finance movement and its various implementations and experiments have existed for decades. However, Bitcoin provided a new tipping point for the movement which led to widespread attention of cryptocurrencies, a subset of Alternative Finance.

Unfortunately, much of the conversation about what cryptocurrencies may enable has been narrower than the solutions and experiments developed within the Alternative Finance movement such as complementary currencies like BerkShares⁵.

This should not discredit the success of cryptocurrency. In the cryptocurrency industry there have been completely new innovations, such as Non-Fungible Tokens (NFTs), in addition to solutions like Automated Market Makers and other tools in Decentralized Finance (DeFi). Some of these have enabled new methods for replacing the functions of traditional financial intermediaries, such as lending and exchange systems.

These are groundbreaking accomplishments with great thought leadership and complex technology.

However, despite ambitious and diverse discussions during crypto's infancy, the conversation has stagnated as many minds have shifted to maximizing or permuting specific ideas within the ecosystem rather than pushing in new directions. There is substantial depth, but there can be radically more breadth to the innovations in the crypto space.

Both the cryptocurrency ecosystem and the alternative finance movement can benefit if part of the thought leadership shifts to incorporating a broader set of tools and goals from the alternative finance movement into the cryptocurrency space. These discussions should fundamentally question how assets are defined, created and utilized, along with developing new methods of contracting and ownership. Again, instead of questioning how cryptocurrency can optimize our current economic system or, we should collectively imagine the economic system we want and design new solutions that are fundamentally different from the status quo. Thinking 'outside the box' is a common suggestion, but with the possibilities enabled by cryptocurrency future progress can follow a completely unprecedented trajectory.

It is also key to note that much of what the cryptocurrency industry has focused on to date will be fundamental infrastructure necessary for implementation of new financial systems, such as exchanges, stablecoins, oracles, and DAO's. Further, even abstract concepts like NFTs may provide fertile ground to experiment with new contract and ownership methods which can be adopted for other asset classes in the future.

Given the facts above the next sections dive into the history of cryptocurrency and selectively chooses topics based on their relevance to discussion of ScriptDole.

V. **Bitcoin**

Bitcoin is a decentralized, permissionless ledger which enables the transaction of value between two parties without the exchange of any personal information or utilization of any intermediaries. The revolutionary technology that enables this is the Blockchain, a series of

⁵ <https://www.berkshares.org/>

'blocks' of data cryptographically linked together so that the history of any Bitcoin 'token' can be traced through every transaction back to the token's origin.

ScriptDole is built on the Radix decentralized ledger network, which does not utilize a blockchain like many other decentralized networks. Radix relies on a unique consensus mechanism called Cerberus. This is crucial as Cerberus enables key features for not only ScriptDole, but for anyone who wants to build their own currency system using the 'blueprints' from the SBI model behind ScriptDole.

It is worth recognizing the role Bitcoin's developments played in history to understand how ScriptDole can and should be structured even if it does not use the same underlying technology. Part of this recognition is also to help frame the conversation and understand that nothing about Bitcoin's 'tokenomics' is inherently necessary for a currency.

A finite number of tokens printed according to a specific plan is not a prerequisite for an alternate finance system, nor is a currency that is marketed as deflationary. These aspects of Bitcoin may have been critical to Bitcoin's success and sustained enthusiasm, but some of these aspects were crucial largely because of Bitcoin's position as the first cryptocurrency to market and may no longer be necessary.

For example, Bitcoin awarded a substantial amount of the total token distribution to early adopters. Part of this may be in part because Bitcoin's security relies on a competitive race to contribute computation power to the network. The high rewards for early adopters was critical to draw in the first 'miners' who power the network.

There is no reason why rewarding early adopters is inherently valuable, and disproportionate early rewards may even lead to some economic ills that society has fought in centuries past. For example, there is the Rule Against Perpetuities in several Western legal jurisdictions which is a means to stop the practice of landed estates. In a similar vein, society should be wary of new instruments that are handing a disproportionate amount of wealth based solely on being the first to own an asset. Excessive rewards for early adopters may also function as a turn-off from many potential adopters and debates about token allotments in cryptocurrency are frequent conversations across crypto communities.

Luckily, because Bitcoin won the public's attention, there are other ways to gain security with an early stage project. For example, ScriptDole inherits the security of the Radix network which is based on "Proof of Stake" - an idea that did not enter the discourse until after Bitcoin started the cryptocurrency movement. This allows development of other financial system alternatives, such as an unbacked universal basic income, in which the supply may increase indefinitely (but with scarcity dictated by time and participant count.) All of this is done without making some of the sacrifices that Bitcoin had to make such as disproportionate rewards for early adopters.

Taking note of this is critical as Bitcoin's distribution could be a design flaw that makes it susceptible to the same type of economic critique which led to its invention. Bitcoin was released during a period of economic turmoil marked by dissatisfaction with government policies such as quantitative easing and bailouts for banks. Bitcoin addressed this by offering a fixed policy of currency distribution enforced by a technological consensus mechanism. This distribution cannot be altered.

However, Bitcoin's price and adoption is still subject to social consensus. A simple question of asking how the market would respond if Satoshi - the creator of Bitcoin - moved any of his bitcoin reveals that Bitcoin may be subject to arbitrary value movements from an unaccountable party. This is not to say that Satoshi is an unaccountable party, but many of Bitcoin's adherents might be just as dissatisfied with Satoshi moving all value from the genesis wallet to Dogecoin via an exchange as they would be if a government inflated its fiat currency. This move, from a single actor, could throw multiple markets out of equilibrium. Both the potential Satoshi or government actions may seem out of line with the ethos that Bitcoiners or citizens expect from those who are leading the social consensus of a currency or the literal printing of a fiat money.

The goal of this paper is not to discredit Bitcoin's achievements or discuss what role it should play in society, but rather to note that power is fluid and accepting a maximalist dedication to any one philosophy could be dangerous and may even undermine the tenets of a specific project or policy. Thus, cryptocurrency enthusiasts should always be concerned not only of attacks to the technical consensus mechanisms, or characteristics of a currency's 'tokenomics,' but also of any social attacks against a particular currency.

Further, the cryptocurrency and alternative finance community should persistently take view of what type of society they want to build and question what type of currency systems will create that vision. This approach is opposed to simply selecting the most preferable system presented by the status quo. If salesmen should always be closing, then cryptocurrency enthusiasts should always be building.

VI. Programmable Ledgers

While Bitcoin broke ground and introduced the world to cryptocurrency, the next major historical event in the course of cryptocurrency's evolution was likely the emergence of Ethereum, another cryptocurrency which also enabled 'programmable' money. Not only is Ethereum a ledger which accounts for the history of every Ether token on the network, it also allows developers to put code on the network and allow any participant to interact with it knowing that the code will always execute as expected.

This groundbreaking development meant that things like automated exchanges could be developed. This replaced even more financial intermediaries, and also opened up an endless conversation about what types of rules we as a society may want to put in place when developing systems of ownership, finance and contract.

Ethereum's growth, and the explosion of competitors plus applications built on Ethereum or other 'Layer-1's' (the protocol level upon which other applications can be built) has demonstrated the market potential for this type of thought. The continued evolution of programmable money and assets should enable more alternative finance concepts to become mainstream and with much lower costs than at any point in history. Additionally, while lowering the costs it also opens the room for experimentation as ideas can be reused or permuted endlessly across the world and within and between communities.

However, in designing ScriptDole and the underpinning SBI model, there are key characteristics which have led to the choice of the Radix network instead of more well known options like Ethereum.

VII. Why Radix

For ScriptDole and SBI to succeed the infrastructure it is built on must meet 5 requirements:

- High transactions per second (TPS)
- Easy for any other party to copy and redeploy the RUBIES model
- Composability with other financial instruments
- Native asset on the network
- Ability to assess if a user is unique

High TPS

High TPS is needed because if ScriptDole is going to be successful it will need to operate within the same network that runs the world's entire financial system. Today, Alipay alone has already hit transaction rates as high as 120,000 TPS⁶, something that no cryptocurrency has yet to achieve on a public ledger. Ultimately, the world's financial system - especially as new asset classes and trading applications are created - will have a TPS rate orders of magnitudes higher than it is today.

The Radix protocol has taken a 'start at the end and work backwards' approach and designed a network from the ground up which should accommodate nearly infinite TPS. This means that the network will not need upgrades while in operation just to meet evolving demand, and that not only should the financial system be able to operate on the system, but the cost of executing a transaction should remain low.

Easy to Redeploy

Second, ScriptDole's goal is to be the Bitcoin of SBI's in terms of starting innovation, but is indifferent as to whether ScriptDole should be the market leader in terms of market share. If ScriptDole starts a conversation that leads to other types of universal basic income systems

⁶<https://www.forbes.com/sites/zennonkapron/2020/05/24/chinas-central-bank-digital-currency-will-strengthen-alipay-and-wechat-pay-not-replace-them/?sh=5ca45df26b69>

that are more accessible or more empowering then the ScriptDole Foundation will consider its efforts a success.

Hence, ScriptDole uses the Radix network, as on the Radix network applications are not uploaded directly to the network by a developer as they are with the Ethereum Virtual Machine (EVM) and other “Layer-1” cryptocurrency protocols.

Rather, a developer publishes a ‘blueprint’ on the ledger then makes a call to the blueprint to create an instance of it on the network. Later, if someone else wants to utilize the developer’s work, instead of copying the code from a repository and then editing, as a developer would need to do with the EVM, the new user can make an on-ledger call to the blueprint and have their own instance of it on Radix.

What this means is that anyone can make their own instance of ScriptDole’s SBI model on the Radix network with extremely minimal work and with the security of knowing that the code will be redeployed in the same exact manner as the original, inheriting the same functionality and level of security.

This should lead to easy adoption of alternative SBI’s for things like local economies, DAO’s, or Network States.

Further, it allows the ScriptDole foundation to be agnostic as to whether the best path forward is a Know Your Customer (KYC) approach or a truly decentralized, anonymous and permissionless approach. The ScriptDole foundation is utilizing a KYC approach not because of inherent desire to be part of traditional financial systems but rather due to the need for an easy way to ensure no one is creating more than a single account. If users could create multiple accounts then the currency would collapse. Outside of ScriptDole’s KYC approach is an open invitation to other parties to make their own permissionless SBI’s from Day One by having easy and reusable blueprints.

Composability

Third, ScriptDole needs to be composable with other financial applications, and must be on a network that offers atomic composability - a requirement for ScriptDole’s disbursement mechanism.

Composability means that every application or resource on the network can work in conjunction with any other application or resource. This leads to an unparalleled level of innovation as anyone can come and experiment with permutations of other people’s work.

Cryptocurrency is only a subset of the alternative finance movement. Over time the number of ideas in the cryptocurrency space will grow and positioning ScriptDole composable with any of these new projects is essential. Hence, ScriptDole needs to be on a platform that allows composability between other components of the ecosystem if it is going to flourish in the context of emerging alternative finance concepts.

Meanwhile, atomic composability means the ability to settle a multi-step transaction as a single transaction.

ScriptDole's dispersal mechanism requires atomic composability as it needs to check the price of ScriptDole, issue new ScriptDole tokens, immediately sell some to an exchange for XRD, then conduct the actual dispersals to individual addresses. All of this needs to be done while using the XRD the component just earned to pay transaction fees on the Radix network. If this application was built on another ledger these would need to be separate and sequential actions taking numerous steps to execute. This decreases efficiency and security. This fractured composability also makes it possible that the cost of the transaction could change part way through the process. A change in price could leave insufficient funds to complete the transaction despite funds already wasted on other legs of the transaction.

Beyond Radix other decentralized ledgers break composability with their scaling mechanisms.

Meanwhile, the design of Cerberus, Radix's consensus mechanism, makes it so that the network is both highly scalable and so that all parts of the network are atomically composable with each other. Thus, ScriptDole's utilization of Radix demonstrates a commitment to the idea that maximalism is bad and that connecting with an evolving alternative finance ecosystem is always preferable. This should also open up more use cases which will further enhance the value of ScriptDole.

Native Asset

A further advantage of Radix is that ScriptDole will be a native asset to the protocol.

For Ethereum and other "Layer-1's", typically only the chain's official asset, such as Ether on Ethereum, is actually native to the protocol. What this means is that other tokens on the network do not inherit the same functionality and security as the native asset and the associated selling points of cryptocurrency security. This is in part why hacks are so common in the decentralized finance space. Rather, on Ethereum and other networks, assets and tokens are created via messaging contracts on top of the ledger which communicates balances between sets of smart contracts. For example, the ERC-20 token on Ethereum is actually a design template, and not an actual token 'minted' by the protocol.

Meanwhile, on Radix, every token is treated by the network the same way that the native XRD token is treated. This means that every token inherits the same security and functionality of XRD.

While extremely beneficial in general this also helps adoption as any wallet or payment system capable of sending XRD or other Radix based tokens should easily have the same ability to send other Radix based tokens. This means that ScriptDole can 'piggyback' on general adoption of the Radix ecosystem and be a click away from use wherever Radix is

accepted. It is even possible for developers to build wallets that prioritize which currency should be spent first in a transaction, e.g. “pay for this transaction with ScryptDole first, then XRD for the remainder.” This is relevant as it means ScryptDole may have an early use case even if its actual exchange value is low. The frictionless power of this design is substantial and allows ScryptDole to benefit with synergies as Radix and projects on Radix gain traction. This also extends to Radix’s single sign-on solution for KYC, Instapass, as any user who gets Instapass verified for any reason can use that verification to register for ScryptDole. The inverse is also true: people in the Radix community may want to champion ScryptDole as a social basic income token is a gateway to going through Instapass and opening the door to other DeFi applications and assets.

Unique Users

Finally, for ScryptDole to function, there needs to be some way to ascertain that a user is not ‘double dipping’ on the network. Each person is entitled to their remittance of ScryptDole, but they should not be allowed to sign up twice and receive more. Hence, there needs to be some method to ensure each user is not repeated elsewhere on the network. ScryptDole solves this by utilizing Radix’s Instapass service, a Single Sign On (SSO) solution for Know Your Customer (KYC) compliance.

As mentioned earlier, the ScryptDole foundation is agnostic as to the importance of KYC compliance, but at this time Instapass is the lowest cost and most ready solution. In the future, as more decentralized identity solutions come into practice, competitors may develop alternatives that are more in line with the public’s demands. However, this is the best go-to-market strategy for ScryptDole. Instapass does not need to be the only solution for ScryptDole either. Other means for registering a user may arise and even the decision tools for authorizing registration methods may move to community governance at some point in the future.

VIII. Complete Decentralization: Local Universal Basic Income & Non-Transferable Badges

The desire for additional implementations of SBI’s by other market participants is not only to spread the word or create economic experimentation, but also because local social basic income systems paired with organizations built from the ground up with Non-Transferable Badges⁷ could be the foundation for decentralized identity. This type of grassroots movement could even create a decentralized solution to replace the Instapass verification for ScryptDole.

Non-Transferable Badges can perform functions such as permanently associating a user’s online identity with anything from academic credentials to DAO acknowledgements or rights to use specific assets. These features can be easily implemented using Radix’s badges, another resource native to the network.

⁷ Non-transferable badges are a more elegant implementation of “Soulbound Tokens.” Badges are a native resource on the Radix network.

When implemented on a local level it is possible for a social basic income system to rely on pre-existing, community level connections for verification for badges rather than using centralized or official entities and institutions like Instapass verification. Thus, a community does not need to have access to wider identity systems to launch their own SBI system based on Non-Transferable Badges.

Further, the distribution of the social income token creates a mechanism to incentivize only allowing valid identities. If the authorizing group or entity allows too many fakes then they are diluting the power of the social token.

Additionally, this is a level at which fraudulent organizations can be easily replaced. If a community organization starts accepting too many fake accounts then it should be easy for an alternate group to launch a new token using the same blueprints from the Radix ledger. Thus, the nature of the free market forces groups to design solutions with their constituents in mind or else the token risks losing users who vote with their feet.

Eventually, the identities and organizations built from the ground up using Non-Transferable Badges and SBI's can lend legitimacy to other organizations and broader or higher levels/hierarchies than the local economy and small groups. Hopefully this can incentivize foundations for global identity systems built on top of decentralized local authorities and also greater empowerment, organization, and collective action systems. All of this can be used not only for economic tools, but also for decentralizing other parts of society (DeSoc).

As an aside, for purely online and anonymous groups, even if a single person two Non-Transferable Badges or income streams in the same community by using different aliases, the commitment to each identity will likely be so substantial that it may not be misplaced to say the holder is contributing enough to that community via different avatars so as to deserve two UBI income streams.

IX. Extra Tokenomics & “Confidence Farms”

The goal of ScriptDole is to take a long term and sustainable approach. Further, many market participants may want to take a long view on ScriptDole as well and the additional ‘tokenomics’ and Confidence Farming are a means to let these participants benefit from their long term confidence or vision for the success of the token. The Confidence Farm also provides liquidity, takes tokens out of circulation (temporarily), and hedges against impermanent loss in trading pools.

ScriptDole aims to do this via Automated Market Makers (AMM)’s, Liquidity Pools for the AMM’s (as used in existing DeFi applications), and a new tool, “Confidence Farms.”

Liquidity pools are a means by which Automated Market Makers (a type of decentralized exchange) maintain liquidity while always offering buying/selling options on the exchange. An

investor deposits both tokens of an exchange pair into its liquidity pool and then earns a fractional share of the AMM's profits earned from trades on the exchange.

In the future, it can be possible for organizations and people to use a new "Confidence Farm" application.

A Confidence Farm is a new type of financial intermediary between the investor and the liquidity pool. A transaction starts with the investor sending tokens to the Confidence Farm along with a pledge of how long they will keep the tokens locked in the liquidity pool. The Confidence Farm then sends the tokens to the liquidity pool and does not let the investor withdraw until the pledged period has ended. If an investor withdraws their tokens early they must pay a breakage fee which is distributed to the remaining Confidence Farm participants. This is akin to fees paid when breaking a Certificate of Deposit with a traditional bank.

In addition to the liquidity pool profits and the breakage profits, investors locked in the Confidence Farm will also earn extra ScriptDole tokens. These will come from the minting which occurs during the dispersal transaction. The ScriptDole component will mint new ScriptDole tokens at a fixed rate. Then, the first priority will be using the tokens to swap for XRD and pay Radix transaction fees. After this a portion of the remainder will be paid to investors locked in the Confidence Farm.

Current considerations for the Confidence Farm only designate a 'portion' of remaining tokens as it allows the Confidence Farm to implement an algorithm which provides exponentially larger rewards (up until a ceiling.) This creates a new problem of "impermanent deadweight loss," in which the component will be minting tokens that are not used to pay transaction fees, nor are they used to reward Confidence Farm investors. To fully utilize these resources ScriptDole's current plan is to have 'impermanent bonuses,' which will distribute the excess token supply to Confidence Farm holders beyond the standard dispersal ceiling. This will only occur until the impermanent supply from prior deadweight loss is gone.

Liquidity Pools and Confidence Farms are not intended to be an aggressive investing strategy, but rather a form of investing that earns modest returns with low risk, similar to other types of assets a charitable organization may hold in the status quo but with the benefit of buttressing the value of a social universal basic income. Thus, their use may more closely resemble the functions of a decentralized Federal Reserve, or traditional assets like bonds and certificates of deposit by partially controlling the supply of tokens at any given point in time and also knowing how much is locked in for liquidity. Another way to view it is an inverted fractional reserve banking system. When banks work with fractional reserves they extend more credit than what they have in reserve, but tools like Certificates of Deposit help the bank remain stable because they the size of the deposits they can rely on. With a constant supply of new ScriptDole tokens, the way faith is built in the system is taking tokens out of circulation and into the confidence farm so that the currency can operate effectively and maintain a high enough price for the component to raise funds and execute the dispersal mechanism.

Hopefully, these efforts lead to a stable currency, even if that stability is not necessarily pegged to a specific asset or traditional currency.

The benefits of this type of investing lean more into the ethos of alternative finance tools like micro lending, rather than ultra-hyped investments in the cryptocurrency industry. Further, if mechanisms like a Confidence Farm can enhance stability, and the monthly distribution is a more equitable economic model, then advocates could reason that in some regards it is more appropriate to price goods in terms of ScryptDole rather than fiat - something most cryptocurrencies have yet to achieve.

X. Hypothetical Value Stories

This section aims to build upon the Bitcoin case study in Section III by offering hypothetical scenarios for why various actors may utilize an unbacked currency.

Ultimately, any real world use case, or assignment of value, is unforeseeable. However, it is likely, especially in early stages, that ScryptDole's value will be found in rewarding behavior that the current economic system fails to incentivize or finding innovative ways to organize, take collective action, and make use of underutilized resources.

When designing solutions to solve the homelessness crisis one of the clear ways to solve it has been to simply give people housing. Hopefully, ScryptDole will find that even without utilization of existing financial resources it will be possible to help stimulate the economy simply by enabling anyone to participate in a ledger system and all of the societal and organizational benefits that come with it. The benefits of ScryptDole can be compounded with local SBI's, and other alternative finance tools like complementary currencies.

Additionally, the general concept of Universal Basic Income is gaining traction in society. This is happening at a time when more investors are concerned about ESG and CSR initiatives, and not just profits. Taking it a step farther, some investors are deliberately looking to invest in solutions based on their return for society, not the investor, all the way to the extent where the 'investors' expect their contributions to slowly lose value over the course of their life. Thus, many of the hypothetical scenarios below rely on not just economic factors for inducing appreciation for the token, but also social pressures. This may even be a more realistic approach to certain markets as the field of behavioral economics has shown that many traditional assumptions about rational actors do not hold in practice.

This section should shed light on how ScryptDole and SBI's can be an incremental solution that circumnavigates government taxation in favor of a system where each individual sets their own market price for what they think a SBI should be. Additionally, ScryptDole should also distribute it in a more cost effective way than many government social programs. The monthly dispersal also does not come with any conditions for the recipients - something that some studies and programs have been found to be a more effective way to provide aid.

A Non-Zero Sum Mindset: Win-Win-Win

A brief note for framing the value stories. By using ScriptDole, or accepting it or holding it, there are benefits to society beyond the economic transaction. Many use cases below assume that the actor gains extra benefit by putting upward price pressure on the ScriptDole currency and increases the true value of people's dispersals. This could be an extremely valuable framework, especially for organizations like anti-poverty charities who may be looking to invest their foundation's financial reserves in a new tool like a Confidence Farm which earns them investment returns while also pushing up the value of their constituents' monthly allotments. A true win-win scenario.

This is also part of why ScriptDole is not aiming to peg the income allotment to a fiat currency, like the GLO universal basic income project on Ethereum (announced several weeks before the V.0.1 version of this paper.) While a non-speculative currency is enticing it could also destroy any incentive to hold it. Transactions on the Radix network will be incredibly efficient. If there is any benefit to using a token like GLO or ScriptDole then a user could cheaply and quickly swap on an exchange between dollars to GLO or ScriptDole and then make the transaction. (With Radix's atomic composability this could even be 1 transaction.) Meanwhile, the merchant receiving the token could also immediately swap back to dollars.

If the point of sale is the only part of the token's value story then it only makes sense to hold at the point of sale. This could cause the currency to collapse. This could apply to many assets on a frictionless decentralized ledger. To solve this ScriptDole aims to be an asset which people desire to hold, and desire to invest in via Confidence Farms as a form of financial and social action, which should help maintain its price.

This is also part of why ScriptDole uses the word Social Basic Income rather than pitching it as Crypto Universal Basic Income. Referring to it as Universal Basic Income could confuse or distract from conversations about UBI that are based in fiat, pegged or secured currencies.

Finally, as a side note, by pegging to 100, ScriptDole enables easy conversations about what people should be entitled to purchase from the free market for basic necessities and easily understand what any purchase is as a percentage of a standard dole. Hopefully this leads to a constant debate and push for rights and higher ScriptDole prices, similar to labor movements like unions or minimum wages.

Value Story One: Young Market Participants

Imagine a community in a developing country is struggling because the national currency is devalued, and dollar pegs are not affordable. Meanwhile, ready alternatives such as Bitcoin are too volatile and costly to transact. Also, participation in these systems assumes that someone has access to some capital or resources to start but the community has been left out of the world's progress and many residents have little if any assets in their possession.

In general the community's economy is not resilient and in a large part it is driven by external aid. Unfortunately, even well intentioned policies have a negative consequence. For example, in current real world developing economies the domestic garment industry is often undercut by donations of used clothes from developed countries. This means no local entrepreneurs can bootstrap a business as they are automatically undercut and a charitable action by the donors destroys value creation opportunities for the recipients.

Finally, various resources are not organized, are underutilized and/or often suffer from the tragedy of the commons. Full employment, much less gainful employment is a distant dream.

Enter ScriptDole.

This first hypothetical closely mirrors the case of El Zonte by recognizing that an underserved and underutilized part of many economies is the youth demographic. Similar to how the elder El Zonte residents rejected Bitcoin it is likely that many people outside the community will not want to hear narratives about how a cryptocurrency can provide people with basic income.

However, certain demographics in the community may be willing to give the opportunity a shot. In this example the teenagers of the community, almost all of whom are unemployed despite their wishes, are the first to adopt ScriptDole as it gives them a meager unit of account to exchange amongst each other.

Later, after the teenagers start using ScriptDole to either trade for other assets, or trade amongst themselves, the elders and other market participants take notice and register for ScriptDole. The elders do not have an immediate use for it but understand that the teenagers organize using this tool and that the adults can use their monthly allotments to pay teenagers to do tasks that would otherwise go undone, such as cleaning the river in the El Zonte case study.

In the community a local farmer does not have much to offer to hire people for services and her time is not optimized as she must split it between growing crops and selling the food. She then signs up for ScriptDole and is able to pay for services from an underemployed teenager who is seeking to accumulate more ScriptDole. This could include something such as tilling part of the field or taking crops to market. It also allows the farmer to use the advantages that money provides in terms of being a representative asset that can be moved across time and space. The farmer has nothing to provide the teenager when planting crops as she has yet to reap the rewards from this time consuming process and traditionally the teenager would have no incentive to help at this stage as he gets nothing in return. However, with the monthly allotment of ScriptDole, the farmer has something for immediate exchange. Further, the farmer also has a decentralized form of credit. The farmer can use a smart contract to commit several months of ScriptDole dispersals to be automatically routed to the teenager's account. This creates a decentralized microcredit system that is not dependent on outside financiers or

centralized organization - it can be a permissionless agreement between two parties about who owns a future asset.

Meanwhile, a local activist is concerned about erosion of soil near the community's river. Continued erosion could endanger the community's meager local food supply. The river is a communal resource that everyone benefits from but has no incentives for individuals to expend personal resources on protection or maintenance. The activist registers for ScryptDole and uses it to purchase equipment to plant trees and plants that will prevent further erosion. Further, instead of pitching every community member on helping the effort, he simply requests community members to register for ScryptDole and donate their tokens to his effort. Any income from this can then be reused to pay teenagers to assist and expand the soil erosion efforts.

As ScryptDole begins to flourish in a circular economy (which can be further enhanced by creating a [Complementary Currency](#) backed by ScryptDole), the users also take note that other parts of the world use the currency and it can be traded with people outside the local community. Further, the local community also becomes aware that the world's demand and perception of ScryptDole influences the value of their monthly allotment as measured against other currencies and assets.

With this knowledge the community realizes that it may be underdeveloped from a 'traditional' Western definition of economic development but that it is rich in culture. This leads the teenagers to become entrepreneurial artists who can make NFT's featuring the community's unique cultural attributes and that they can make the NFT only tradable with ScryptDole. This brings in income from the NFT sale and royalties while also increasing demand for the ScryptDole token . This buttresses the price and once again makes the monthly allotments even more valuable to the community. This effort also has the added benefit of making immutable records of cultural traditions that may otherwise have slowly gone extinct as market forces force assimilation into global norms.

Also, in this case study regarding teenagers, the role of ScryptDole could play a profound role in increasing financial literacy at a young age. Not every person in the world is blessed with an allowance based on what they were born into and many people may not really engage with money till later stages of adolescence or even adulthood. By providing youth with ScryptDole it could teach the value of a monthly budget or add perspective on how much the true cost of something is (as everything will be relative to the 100 unit basis of the personal monthly allotment.)

The goal of this case study is to demonstrate the the first chapters of ScryptDole's history may not be about getting the modern world onto a high value universal basic income system where the tokens are priced high enough for trade in advanced economies, but rather that small and underserved communities may benefit from having access to a ledger regardless of what the unit of account is. Once these roots take hold the currency can naturally grow in importance over time.

Value Story Two: Corporate Social Responsibility

Flipping the script (pun intended), this value story turns to how and why large market participants in an advanced economy may want to place value in ScryptDole once it begins to gain traction and paints a rosy picture of what the future may hold.

Many corporations are focused on being good corporate citizens. Some corporations even have charitable foundations and many are in a rush to demonstrate their commitment to uplifting the communities around them and promoting initiatives such as environmental sustainability.

However, CSR is not a metric that is easily quantifiable nor are there universal standards for what it means to be a good corporate citizen.

Utilization of ScryptDole can be an easy and clear way for corporations to send a signal and demonstrate that they support an equitable economy for all. This can be done in addition to serving other interests as incorporating the token into business practices can create price support for the token's value - enhancing the impact for recipients - and also broadening the adoption of the currency so there are more network effects.

If a corporation decides to accept ScryptDole as a form of payment for its goods and services then this is a clear sign that the company believes that a universal basic income has utility and is a worthwhile distribution of economic power. The corporation will also be able to set its own exchange rate for the value of ScryptDole. This exchange rate creates a new dynamic in which a company can use its market position as leverage for social good. Hopefully this will create a race to the top amongst corporations who want to attract customers by demonstrating goodwill with competitive ScryptDole redemption rates, and in aggregate the decentralized price setting by corporations can be a huge factor in determining the market price for ScryptDole.

Beyond accepting ScryptDole corporations can also showcase the scale of their commitment. Many people are questioning whether pure pursuit of shareholder value is the appropriate purpose of a corporation or if the world should adopt more sustainable forms of capitalism. If investors judge corporations in part by adoption of ScryptDole then the future may include earnings reports that feature not only traditional revenue streams but also showcases how much ScryptDole they have brought in and how much they are holding on their books. A further commitment to lock the holdings in a Confidence Farm can be another signal of the company's commitment. All of this is also done while holding an asset that itself is valuable rather than a utopian vision of pure charity. In an idealistic world some corporations may even opt to 'burn' the tokens so they are permanently taken out of circulation and encourage people to think about Earnings Per Share and "Burnings Per Share" when measuring the company's success and role in society.

Value Story Three: Decentralized Direct Cash Payments

It is unavoidable that some people will register for ScryptDole solely to immediately sell their monthly allotment.

However, this does not wholly defeat the purpose of the currency.

Several studies and experiments have shown that direct cash payments are more effective than structured aid programs or money with strings attached⁸. These often empower people not only to make choices, but to direct the funds towards issues that are going to increase their economic empowerment and make them less reliant on aid. For example, in Universal Basic Income pilots many cases have shown that people are actually able to get better jobs over the course of the pilot because the UBI allowed them to cover key expenses without committing excessive amounts of time to their prior, low paying jobs. By making them less dependent on the pay from low income jobs people were able to take time off and apply for new jobs to optimize their opportunities.

Further, direct cash payments also enable recipients to spend the money on more local goods and services rather than structured aid which may have the funds flow to external actors. This further enhances the local economy, driving compounding effects in situations like the first Value Story and creating more economic empowerment and resilience.

What this means is that a free rider may immediately exchange their allotment for another token but also that anyone who is purchasing from an exchange is boosting the value of the dispersals and directly injecting cash into the lives of those who need it most. This also does it in one of the most efficient ways possible by using the ScryptDole component instead of traditional mechanisms such as government taxation and distributions or nongovernmental organizations, who despite their best intents may have many inefficiencies, problems with cross border payments, or high overheads including things like marketing expenses to raise funds to donate laer.

Hence, while the token may experience volatility, and in its early days substantial downward pressure, the money 'lost' by people purchasing a token that may temporarily decrease in value is still providing a valuable contribution to society and contributing to the narrative of ScryptDole. Additionally, this creates an incentive unique from other volatile cryptocurrency assets as price swings in those investments are a pure economic loss or lopsided exchange with another well off party in the crypto ecosystem.

Value Story Four: Perks

While ScryptDole is in its infancy it may serve the currency well for business to offer special perks for ScryptDole instead of pricing all of its goods or services in the currency.

⁸ <https://www.theatlantic.com/international/archive/2015/09/welfare-reform-direct-cash-poor/407236/>

For example, a fast food chain can offer an upgrade from a medium to large soda if the customer pays a small additional fee in ScriptDole. Or it can be something that is only available via ScriptDole purchases, such as a shot of a specialty syrup in your latte.

Both of these actions allow the business to offer an inconsequential good to customers in exchange for ScriptDole and potentially at a better value for the consumer than if they had converted the ScriptDole to another currency and then made the payment.

Also, as mentioned earlier, it should be easy for Radix wallets to accommodate features like “pay for this with ScriptDole first, then pay the remainder with X currency.” This means that an example like the special syrup or soda upgrade is not a complex interaction for the user - they simply pay with their Radix wallet of choice and the network will exchange the appropriate assets behind the scenes. Further, this is not a radical concept for companies. Many businesses currently ask for charitable donations at the point of sale and often with the option to round up to the nearest dollar. This same concept can be applied to ScriptDole payments.

Additionally, if a business wants to accept ScriptDole it could even offer an option to round down the cost of a purchase. For example, if a cup of coffee costs \$3.50, the cafe could offer a customer the option to round the cost down to \$3 if they pay \$0.25 of the difference in ScriptDole. This means that the business is honoring ScriptDole at 2x the market rate, as the customer would have needed to convert twice as much ScriptDole to dollars on an exchange to make the full purchase in cash. This extends the business’ ability to exert leverage for a positive effect on ScriptDole’s value.

Value Story Five: Food Waste & The Hungry

Food waste and hunger are both problems in Western countries which is a bit of a paradox and shows that economics is often the cause of scarcity rather than the actual means of production. If society could do a better job of organizing around this issue then the two problems could solve each other.

For example, a grocery store could sell food that is about to spoil in exchange for ScriptDole. This food would be wasted or discounted anyways, so opting to exchange it for something, even an unpegged social basic income token, would be better than nothing. Also, the nature of the ledger and accounting system could help organize this process more than if the grocery store just aimed to give it away.

Similar to the first value story, the unit of account can help people organize. Examples could include encouraging people to come at the right times to exchange ScriptDole for food, or enabling other parts of the ecosystem necessary to distribute the food before it is wasted, such as getting otherwise unemployed people to deliver the food to a food co-op for a share of the ScriptDole gained from the transactions.

Value Story Six: Loyalty Point Programs

Many businesses have loyalty programs, and some are extremely complex such as those for banks, airlines and travel agencies.

A unique characteristic about these programs is that they have already trained consumers to think in terms of points which are wildly disproportionate to the scale of units for normal currency. For example, an airline loyalty point could be worth a fraction of a cent, but people still seek it out.

The scale at which loyalty points operate can open a window for business to offer redemption programs that make sense and resonate even if the ScriptDole:Loyalty Point ratio is extremely disproportionate. For example, when thinking of the links between dollars, points and ScriptDole, it would not be too far from today's consumer mindset to offer a redemption rate of something like 10,000 Loyalty Points for 100k ScriptDole, or a real world value of \$10.

This means that if a loyalty program wanted to gain a competitive edge against other loyalty networks it could offer better exchange rates for ScriptDole. This is especially valuable upon a close examination of programs like airline loyalty points as they are often a large network of different businesses working together to offer bundled services or connected loyalty programs.

The key point of this value story is not that ScriptDole needs to be exchangeable for points and is actually intended to show that adoption is possible in Western countries even with an extremely low exchange rate between ScriptDole and fiat currencies.

Value Story Seven: Charity Pegging & The SBI Basket

ScriptDole can allow charities to broadcast what ScriptDole price the market needs to maintain for the charity to be self sufficient.

For example, a charity fighting homelessness can announce that it can house all of its constituents if the constituents can sell their ScriptDole tokens at a rate of \$5/token (hypothetical number.) Multiple charities can do this, and this creates a strong societal narrative for why companies and individuals should provide price support that make these charitable operations sustainable without traditional charitable fundraising.

Meanwhile, society can also debate what the value of the monthly allotment should be in terms of what type of purchasing it enables.

It has been established in many studies that the Consumer Price Index is not reflective of the inflation rates that low income workers experience. Prices of electronics may be going down, but this does not help someone who does not have a basket that includes those items because they do not earn enough to have that level of disposable income. Meanwhile, prices for

essentials like food, housing and healthcare may be rising faster than the discretionary items, and these can make up all or most of a low income earner's spending basket.

ScriptDole can introduce a new debate about what type of basket the monthly allotment allows a recipient to purchase.

For example, many activists may push for a price level that allows the purchase of all essentials: food, housing, medical assistance, transportation, and minimal access to the internet so they can fully participate in society. If the SBI basket is something people can actually rely on, then as with the direct cash payments value story it is possible for the recipients to go out and seek better employment opportunities and make better use of their abilities. Additionally, society may benefit even more if people begin doing the work they want to do, rather than just the work that they must do to survive. With more and more automation and mass production there will need to be a shift to in the type of work society demands, and this is in parallel to the adoption of technologies like the metaverse that should enable more dedication to creating leisure experiences.

Value Story Eight: Unions

Unions have historically contributed to the broader labor movement, and not just representation for their members in bargaining against specific employers. Unions can continue to carry this torch by pressuring employers to adopt strong pro-ScriptDole policies, or taking actions like storing union treasuries in a Confidence Farm.

The end goal of this advocacy is not just a better ScriptDole allotment for their members, but it also creates more demand for the products produced by the union members, as the broader society will have more purchasing power if they have a strong SBI economic base.

Value Story Nine: The Reverse Domino Theory

In the Cold War the West was concerned that if one country turned communist, then it was more likely the spread would continue and other countries would fall under the specter of the ideology.

The opposite effect may come from economic empowerment via ScriptDole. As the price of ScriptDole rises to lift the most disadvantaged market out of poverty, then the movement to improve the lives of the next rung of people will become the priority. Thus, while the movement may begin small and in actions like a remote village in an undeveloped country it will hopefully start to snowball and gradually empower more and more people in markets of increasing degrees of sophistication and complexity.

XI. Problems

I. Why use it as a store of value instead of just cashing out each month & swapping for it at the moment of purchase when there is a benefit to using the token?

Given the frictionless nature of Radix this is likely a problem all currencies and tokens may run into if they have enough liquidity.

However, Complementary Currencies could make it hold value, as someone needs to 'break' from the community before sending to an exchange. This means innovation could be used to harness value within a circular economy and allow economic capabilities to grow with some degree of separation from the world⁹

More work needs to be done in this area, but it is not a 'gravity problem' or complete obstacle to the implementation of an SBI.

XII. Competitors

- A. There are several cryptocurrency UBI competitors, but utilization of the Radix network, and the unique tokenomics paired with the goal of turning the token's use into a social movement separates ScryptDole from other solutions in development.
- B. More work needs to be done to learn from the work these projects have undertaken.
- C. Finally, in the end competition is good.

XIII. Next Steps

- A. This Whitepaper will need further revisions as the idea is further fleshed out.
- B. Before implementing the component and currency the goal is to create a non-profit organization for the token. The component will act as a smart contract where administration executes autonomously and the non-profit will have no ability to change certain features after it is launched. However, some features will remain under the non-profit's control, such as the ability to authorize new KYC providers as the project should not be dependent on the continued existence of external services like Instapass. Similarly, connecting the Confidence Farm to multiple exchanges can take continued work. All of the levers of control may also move to a DAO or on-ledger governance as the Radix ecosystem matures.
- C. The ScryptDole component & possible Confidence Farm needs to be coded. Some Radvocates have already volunteered.
- D. It is also ideal for the component to be audited before going live.
- E. There will be no ICO, or premine to raise funds. There is no special token allocation for the non-profit foundation, the founders, or any person or entity. The only means to earn tokens will be through the monthly dispersals of 100 tokens per individual or through the Confidence Farm.

⁹ See Bad Samaritans book.